



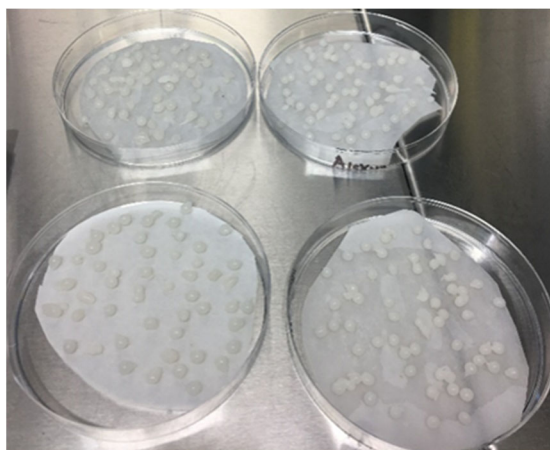
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MATERIALS & REAGENTS

ENCAPSULATION/DEHYDRATION- VITRIFICATION

1. Mix the algae (concentration 10^5 for larger algae to 10^7 for smaller algae) with the sodium alginate mixture (**solution 1**) and Sucrose (**Solution 4**).
2. Then drop gently with a Pasteur pipette into the CaCl_2 (**Solution 2**) to form the beads and allow the beads to equilibrate 60 minutes.
3. Transfer the beads to a sterile petri dish to dry. Let them dry for an hour in a laminar air-flow chamber over filter paper. Then take the beads into criovials, mix them with the cryoprotectant (**solution 5**). After 15 minutes plunge the criovials into liquid nitrogen.
4. To thaw the criovials plunge them into 35°C water for 2 minutes.
5. Then take the beads into a 0,3 M sodium citrate (**solution 3**). Allow the beads to dissolve for 60 minutes.
6. Then take the solution into fresh culture medium and put it into semi dark conditions for 24 hours.
7. Elapsed that time observe the cells under the microscope for qualitative viability assessment /cell fitness.
8. Incubate in normal culture conditions.



(1) **Sodium Alginate:** (sodium alginate 7 g + culture medium 100ml) and shake gently.

(2) **CaCl_2 100mM in culture media:** CaCl_2 11,09 g + distilled water 1 L) into a long test tube

(3) **Sodium Citrate 0.3M:** Sodium citrate 1,54 g + culture medium 20 ml

(4) **Sucrose 0.5M**

(5) **Cryoprotectant** selected in culture media

QUALITY CHECK POINTS POST VITRIFICATION:

Cell density/time

Cell fitness (morphological exam, motility, division)

Fluorescence in vivo

BIBLIOGRAPHY FOR FURTHER CONSULT:

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